Ben Yang

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4th-year PhD student (expected Feb 2028) building real-time XR (AR/VR/MR) interfaces and 3D Gaussian splatting (3DGS) systems for human-robot interaction (HRI) and robot teleoperation

Education

Columbia University, New York, NY | GPA: 3.99/4.00 — Sept 2023 - Feb 2028 (expected)
PhD in Computer Science
Advisor: Steven Feiner

Columbia University, New York, NY | GPA: 3.95/4.00 — Jan 2021 - Dec 2022
M.S. in Computer Science: Vision, Graphics, Interaction, and Robotics Track

Rutgers University, New Brunswick, NJ | GPA: 3.89/4.00 — Sept 2016 - May 2020
B.S. in Computer Science, Math, Cognitive Science; Minor in Philosophy
Summa Cum Laude | Highest Honors in Computer Science

Research Experience

Toyota Research Institute, *Research Intern, Robotics Team* — May 2026 - Sept 2026

Worked on a real-time 3D Gaussian splatting (3DGS) scene memory system for immersive robot teleoperation on the Apple Vision Pro:

- Built a real-time egocentric 3DGS reconstruction pipeline that turns the robot's head and wrist stereo RGB-D streams into world-space Gaussian splats without per-scene optimization, removes robot and background geometry, and keeps occluded foreground objects in a world-voxel memory
- Streamed only scene changes to an untethered Apple Vision Pro, letting operators view the workspace from any viewpoint while the reconstruction updates 5.25 times per second
- Ran a 16-participant within-subjects user study of teleoperated robotic manipulation against a stereoscopic camera-view baseline: 1.78× the odds of completing a manipulation attempt, lower reported workload, higher usability, and preferred by 13 of 16 operators
- Benchmarked on held-out external-camera recordings against three baselines: feed-forward DepthSplat, offline 4DGS, and a 4D Gaussian transformer

Computer Graphics & UI Lab, *PhD Candidate, Research Assistant* — Sept 2023 - Present

Working with Professor Steven Feiner on XR interfaces for HRI and robot teleoperation, real-time 3DGS, and XR toolkits (funded by the Future Manufacturing in Robotics and Columbia-Dream Sports AI grants):

- Designed and built Delegate the Details, an XR interface and robot planner for authoring assembly constraints for human-robot collaboration, evaluated in a 20-participant study and offline with a simulated UR5 robot (IEEE ISMAR 2026, IEEE VR 2026)
- Developed a real-time feed-forward 3D Gaussian splatting pipeline on Meta Quest 3 that digitizes real objects in-headset in 2-4 seconds as interactive, reality-linked proxies (IEEE VR 2026, Best Demo Honorable Mention)
- Conducted a 15-participant user study comparing the benefits and drawbacks of permanent and temporary proxies of objects for bimanual selection and manipulation in VR (IEEE VR 2025)
- Built VHard, an XR kinesthetic-rehearsal system for bouldering on a MoonBoard digital twin with individually 3D-scanned holds, and ran a climbing-gym user study
- Created a fluent C# DSL for spatial and hierarchical message routing on the MercuryMessaging toolkit, evaluated with human developers and LLM-based AI coding agents
- Built an XR sand table for a DARPA-funded project that lets many officers collaborate on mission planning in XR, using gaze and hand tracking for menuing on Quest 3 and Varjo XR-3

Computer Graphics & UI Lab, *Research Assistant* (20 hrs/wk) Sept 2021 - Aug 2023

Worked with Professor Steven Feiner for the AntARctica & MercuryMessaging projects:

- Manipulated and cleaned terabytes of radar data from Antarctica ecological projects to construct a Unity VR application for easy data viewing and visualization
- Included networking in AntARctica allowing multiple users to interact with the scene together
- Built a Matlab radar to OBJ converter that compresses the OBJ to avoid Unity loading errors
- Released MercuryMessaging 2.0, an open-source routing and event handling library for Unity that speeds up VR app and user study development up to two-fold, including networking APIs

Software Systems Lab, Columbia U., *Research Assistant* (15 hrs/week) Jan 2022 - May 2023

Worked with Professor Junfeng Yang for the Cerberus project:

- Modified Google Electra to take source code instead of natural language data and trained models for bug detection since natural language does not capture data / control flow of programs
- Pretrained various BERT, Grace, SPT-Code, etc. deep learning models from scratch and then fine-tuned on Java Defects4J dataset to compare against Cerberus for an ablation study
- Achieved a 70% accuracy on detecting security vulnerabilities in C code (the Devign dataset) using natural language processing (NLP) based off Google Electra

AWS CodeGuru, Amazon, *Applied Scientist Intern* (40 hrs/week) May 2022 - Sept 2022

- Created AST parser for Java, Javascript, Python to be used in ML CodeGuru models, and used parser to generated gigabytes of data from private Amazon and public Github codebases
- Wrote commit history dataset builder for any Github / Amazon repository, reduced dataset creation time by 100x
- Used dataset to build graph-based bug fixing models for CodeGuru, allowing CodeGuru users to automate code reviews and detect more security vulnerabilities

Alexa AI, Amazon, *Applied Scientist Intern* (40 hrs/week) May 2021 - Sept 2021

- Leveraged multilingual language models such as mBART and XLM-r for semantic parsing tasks
- Annotated and parsed 2.5 million data utterances, and used machine translation on French, Hindi, and Italian data for training
- Achieved an 80% accuracy on English and 60% accuracy on French, Hindi, and Italian data

Combra Lab, Rutgers University, *Research Assistant* (10 hrs/week) Spring 2019

Worked with Professor Konstantinos Michmizos for the Astrocytes project:

- Recreated SNN with CNN-inspired architecture following Diehl et al. 2015 STDP Paper, built neural layers that resemble Astrocyte brain cells for improved MNIST digit recognition accuracy

Honors & Awards

Best Research Demo Award (Honorable Mention), IEEE VR 2026 Mar 2026

Winner, Columbia-Dream Sports AI Grant Sep 2025

Greenwoods Fellowship, Columbia Engineering 2024

Winner, EGSC Professional Development Scholarship Oct 2024

Grand Prize, AGU Michael H. Freilich Student Visualization Competition Nov 2022

Honors College Scholar, Rutgers University Sept 2016 - May 2020

Cap & Skull Senior Honors Society, Rutgers University Sept 2019 - May 2020

hackNY Fellow, New York, NY May 2018 - Sept 2018

Dean's List, Rutgers University Sept 2016 - May 2020

MANUSCRIPTS UNDER REVIEW

First-author manuscript submitted to the 2027 IEEE International Conference on Robotics and Automation (ICRA), under double-blind review (title withheld during review).

First-author manuscript submitted to the 2027 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), under double-blind review (title withheld during review).

First-author manuscript submitted to the 2027 ACM CHI Conference on Human Factors in Computing Systems (CHI), under double-blind review (title withheld during review).

PREPRINTS

B. Stotz, N. Wang, D. Nogina, C. Zhang, J. Yin, A. Huang, **B. Yang**, J. Li, J. Salzman, S. Feiner, and S. Sellán, "PointGrade: Geometric Priors for Grading MoonBoard Problems," arXiv preprint arXiv:2609.17770, 2026.

PUBLICATIONS

Z. Li, X. He, H. Chen, C. Zou, S. Bai, **B. Yang**, M. Wu, J. Ledner, Y.-J. Cheng, A. Yamauchi, D. G. Turakhia, S. Feiner, and P. Sajda, "Augmenting Human Performance with an XR Agent Learning from Online Behavior and BCI Evidence," 2026 ACM Symposium on User Interface Software and Technology (UIST), Detroit, MI, USA, 2026 (to appear).

B. Yang, X. He, C. Zou, J. Liu, B. Tversky, and S. K. Feiner, "Delegate the Details: Flexible Assembly Constraint Authoring for Collaborative XR," 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Bari, Italy, 2026.

Z. Li, X. He, M. Wu, Z. Tong, H. Wei, **B. Yang**, S. Feiner, and P. Sajda, "SwEYEpinch and Beyond: Exploring Intuitive, Efficient Text Entry for Extended Reality via Eye and Hand Tracking," 2026 ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026, pp. 1-27, doi: 10.1145/3772318.3791820.

B. Yang, S. Liu, K. Natarajan, and S. K. Feiner, "Fast Digitization in XR of Reality-Linked Gaussian-Splatted Proxies," 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), **Best Research Demo Award (Honorable Mention)**, 2026, pp. 1429-1430, doi: 10.1109/VRW70859.2026.00389. [Demo video]

B. Yang, X. He, C. Zou, J. Liu, B. Tversky, and S. K. Feiner, "Bimanual XR Specification of Relative and Absolute Assembly Hierarchies for Teleoperation," 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Poster, 2026, pp. 1033-1034, doi: 10.1109/VRW70859.2026.00191. [Presentation video]

B. Yang, X. He, C. Zou, J. Liu, B. Tversky, and S. K. Feiner, "Virtual Reality Task Guidance Through Relative 6DoF Pose Specification," 2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), Daejeon, South Korea, 2025, pp. 901-902, doi: 10.1109/ISMAR-Adjunct68609.2025.00247.

B. Yang, J. Liu, X. He, B. Tversky and S. Feiner, "Permanent Proxies for Bimanual Selection and Manipulation in VR," 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Saint Malo, France, 2025, pp. 1330-1331, doi: 10.1109/VRW66409.2025.00311. [Presentation video]

B. Yang, X. He, J. Li, C. Elvezio and S. K. Feiner, "An XR GUI for Visualizing Messages in ECS Architectures," 2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), Bellevue, WA, USA, 2024, pp. 640-641, doi: 10.1109/ISMAR-Adjunct64951.2024.00189. [Presentation video]

J. Salzman, J. Li, **B. Yang** and S. K. Feiner, "VHard: An XR UI for Kinesthetic Rehearsal of Rock Climbing Moves," 2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), Bellevue, WA, USA, 2024, pp. 638-639, doi: 10.1109/ISMAR-Adjunct64951.2024.00188. [Presentation video]

A. Boghosian, S. I. Cordero, C. Elvezio, S. Sanchez-Zarate, **B. Yang**, S. Guo, Q. Ashikin, J. Salzman, K. Tinto, S. Feiner, and R. Bell, "Augmented Reality and Virtual Reality for Ice-Sheet Data Analysis," IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, Pasadena, CA, USA, 2023, pp. 52-55, doi: 10.1109/IGARSS52108.2023.10283077. [Presentation video]

Conference Presentations

Paper talk, "Delegate the Details: Flexible Assembly Constraint Authoring for Collaborative XR," IEEE ISMAR 2026, Bari, Italy — Oct 2026

Research demo, "Fast Digitization in XR of Reality-Linked Gaussian-Splatted Proxies," and poster, "Bimanual XR Specification of Relative and Absolute Assembly Hierarchies for Teleoperation," IEEE VR 2026, Daegu, South Korea — Mar 2026

"Virtual Reality Task Guidance Through Relative 6DoF Pose Specification," IEEE ISMAR 2025, Daejeon, South Korea — Oct 2025

Poster, "Permanent Proxies for Bimanual Selection and Manipulation in VR," IEEE VR 2025, Saint-Malo, France — Mar 2025

Demos, "An XR GUI for Visualizing Messages in ECS Architectures" and "VHard: An XR UI for Kinesthetic Rehearsal of Rock Climbing Moves," IEEE ISMAR 2024, Bellevue, WA — Oct 2024

Work Experience

Datadog, NYC, *Software Engineering Intern* (40 hrs/week) — May 2019 - Sept 2019

- Developed a Python service to aggregate millions of Slack and HipChat notifications for enterprise customers, allowing users to see developer statistics for downed microservices
- Worked with Kubernetes, Cassandra, Kafka, and containerization / data wrangling tools

MakeSpace, NYC, *Software Engineering Intern* (40 hrs/week) — May 2018 - Sept 2018

- Developed Item Tagging Widget, an in-house machine learning model used to tag about 1 million photographs of stored items, using TensorFlow and other computer vision APIs
- Wrote a data analysis report on the volume of stored items using D3.js and React

XLP Capital, NYC, *Software Engineer* (40 hrs/week) — March 2018 - April 2019

- Built data source integrations (loading and analysis) for Interactive Brokers, Bloomberg, CCXT, Quandl finance APIs
- Developed trading platform GUI in J, allowing for resizing, tabbing, Excel functionality and handling of 2GB sized data using JQT Framework

MetLife, Software Engineering Intern — May 2018 - Sept 2018

- Developed Go Multithreading Tester App that stress tests MetLife databases for correct multithreading and functionality
- Worked with MongoDB and Java to develop automated derivative transaction builder and library for millions of transactions
- Used Python to download & sort 100GB+ US agricultural and climate data

Teaching Experience

COMS 6173: VR & AR Seminar, Columbia University, *Head TA* — Fall 2025

- Managed 8 research projects for class student groups
- Held weekly office hours, graded exams and homework

COMS 4172: 3DUI and AR, Columbia University, *Head TA* — Spring 2024

- Created 3 mobile AR and headset VR assignments for MS and undergraduate students
- Held weekly office hours, graded exams and homework

COMS 4771: Machine Learning, Columbia University, *Teaching Assistant* Summer 2021, Fall 2021

- Held weekly office hours, graded exams and homework

Computer Science Dept., Rutgers University, *iLab Assistant* Spring 2019, Fall 2019, Spring 2020

- Tutor at CS Lab for Data Structures, Algorithms, Computer Architecture, Operating Systems

First-Year Interest Group Seminar, Rutgers University, *Instructor* Fall 2018

- Taught an accredited college course "Exploring Computer Science" with 10 activity-based lesson plans to a diverse group of 25 first-years, including topics such as Arduinos & ML/AI
- Covered the history of CS and current diversity issues in both industry & academic job markets

Math 477: Probability Theory, Rutgers University, *Grader* Fall 2017

Peer Review

Reviewer: ACM CHI (2024, 2025, 2027), ACM VRST (2024), IEEE ISMAR (2024)

Leadership & Service

Student Alliance of Computer Scientists, Rutgers University, *Vice President* Sept 2016 - May 2020

- Built a strong Rutgers tech community by establishing monthly speaker series with 10+ speakers reaching a 300+ student audience, partnered with other tech orgs such as STEAM and WiCS
- Acquired thousands of dollars of sponsorship for HackRU bianually
- Headed RUnite CS Ambassadors with 15 student developers, visiting 20+ middle and high schools, reaching 1500 students, mentored 5 students for career and academic advice

Quantitative Finance Club, Rutgers University, *President* Sept 2017 - May 2020

- Orchestrated and taught 14 advanced technical presentations involving trading models, finance, Black-Scholes, and computer science theory
- Reached and helped 100+ members regarding technical and practical knowledge for interviews and career search in addition to weekly mock interviews

Rutgers Enactus, Rutgers University, *Veterans 4 Warriors PM* Sept 2017 - May 2019

- Worked with a team of student entrepreneurs to coordinate intensive ambassador program for marketing and awareness of Veterans' health
- Built relationships with 20+ student and administrative organizations, planned Veterans events, tabling activities, and social events with clients

Petey Greene, Edna Mahan Correctional Facility, *Tutor* Sept 2018 - May 2019

Technical Skills

- Languages: Python, C#, C++, Swift, Java, JavaScript
- XR & graphics: Unity (OpenXR, Meta XR SDK), 3D Gaussian splatting pipelines (feed-forward reconstruction, CUDA rasterization, TensorRT), visionOS (Swift/Metal), Meta Quest 3/Pro, Apple Vision Pro, HoloLens 2, Varjo XR-3
- Robotics & ML: ROS 2, MoveIt, MuJoCo, UR5e, PyTorch, TensorFlow